



# Influence of antibody–drug conjugate cleavability, drug-to-antibody ratio, and free payload concentration on systemic toxicities: A systematic review and meta-analysis

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## Abstract

While in theory antibody drug conjugates (ADCs) deliver high-dose chemotherapy directly to target cells, numerous side effects are observed in clinical practice. We sought to determine the effect of linker design (cleavable versus non-cleavable), drug-to-antibody ratio (DAR), and free payload concentration on systemic toxicity. Two systematic reviews were performed via PubMed search of clinical trials published between January 1998–July 2022. Eligible studies: (1) clinical trial for cancer therapy in adults, (2)  $\geq 1$  study arm included a single-agent ADC, (3) ADC used was commercially available/FDA-approved. Data was extracted and pooled using generalized linear mixed effects logistic models. 40 clinical trials involving 7,879 patients from 11 ADCs, including 9 ADCs with cleavable linkers ( $N=2,985$ ) and 2 with non-cleavable linkers ( $N=4,894$ ), were included. Significantly more composite adverse events (AEs)  $\geq$  grade 3 occurred in patients in the cleavable linkers arm (47%) compared with the non-cleavable arm (34%). When adjusted for DAR, for grade  $\geq 3$  toxicities, non-cleavable linkers remained independently associated with lower toxicity for any AE ( $p=0.002$ ). Higher DAR was significantly associated with higher probability of grade  $\geq 3$  toxicity for any AE. There was also a significant interaction between cleavability status and DAR for any AE ( $p=0.002$ ). Finally, higher measured systemic free payload concentrations were significantly associated with higher DARs ( $p=0.043$ ). Our results support the hypothesis that ADCs with cleavable linkers result in premature payload release, leading to increased systemic free payload concentrations and associated toxicities. This may help to inform future ADC design and rational clinical application.

**Keywords** Antibody–drug conjugates · Cleavable and non-cleavable linkers · Payloads · Drug to antibody ratio · Systemic toxicities · Meta-analysis

## 1 Introduction

Antibody–drug conjugates (ADCs) are monoclonal antibodies connected to a cytotoxic agent known as the payload via a chemical linker. It was hoped that ADCs would be “magic bullets,” delivering high-dose cytotoxic chemotherapy directly to cancer cells without affecting surrounding normal tissues. However, this has not borne out in clinical practice. Though many factors affect toxicity, the toxicities of currently approved ADCs appear to be driven primarily by premature release of the payload into the bloodstream by the linker, by an excessively prominent bystander effect [1], or even payload released by the lysed tumor cells [2].

ADC linkers can be divided broadly into two groups: cleavable and non-cleavable. Cleavable linkers such as hydrazone, disulfide, or peptide linkers rely on physiologic factors

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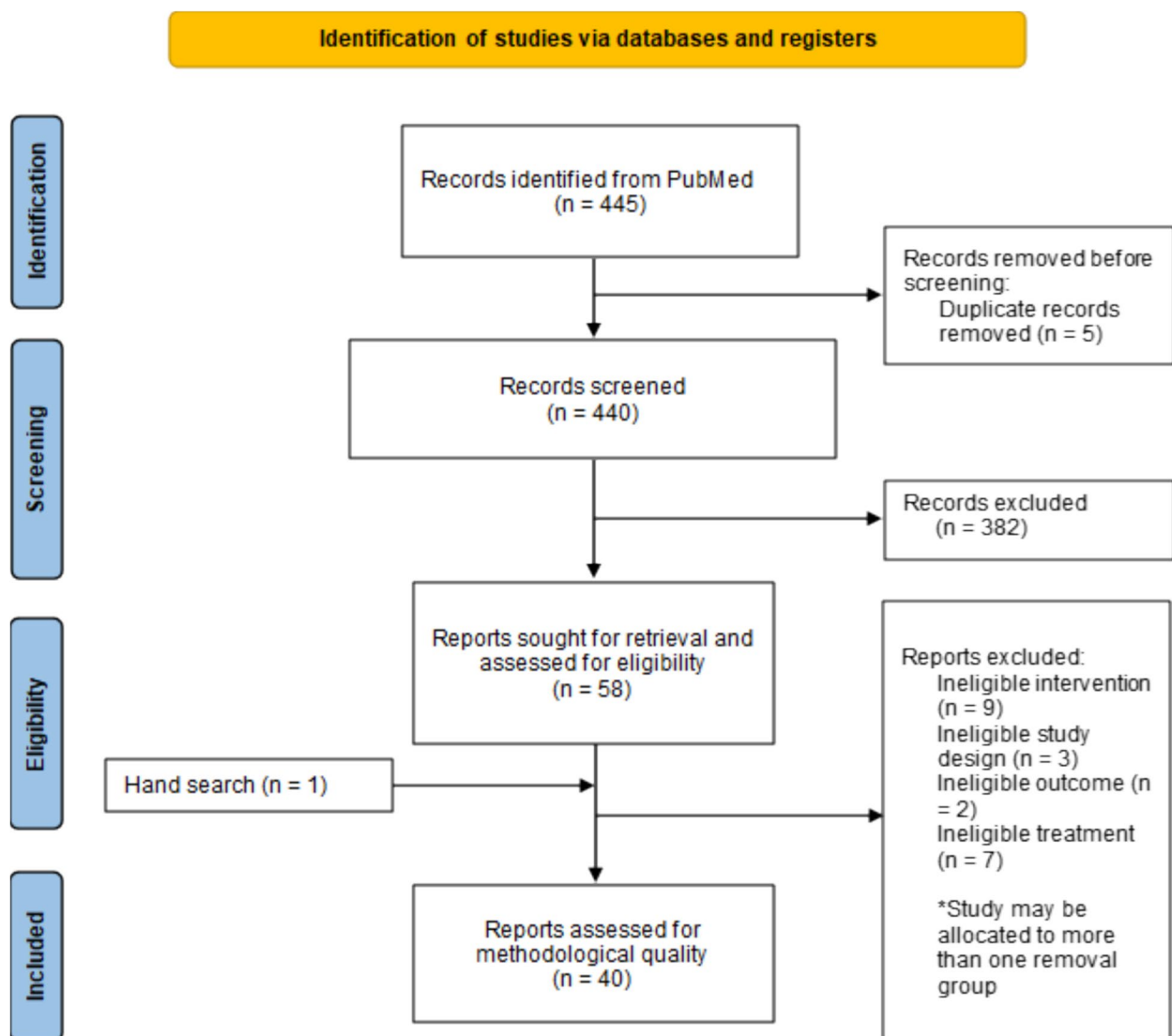
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(i.e., cathepsin, glutathione (GSH), and low pH) within the cell to cleave the linker. Because these conditions can occur independently of antigen internalization, cleavable linkers are often less stable in the blood, resulting in various off-target effects [3]. In contrast, non-cleavable linkers, such as the thioether or maleimidocaproyl linkers, require internalization by the target cell, so that the antibody, rather than the linker, can be degraded by the lysosome before the drug is released. This latter mechanism does not produce efficient bystander killing and thus results in lower toxicity profiles [4]. For these reasons, other novel linkers, including conditionally released linkers, are currently in rapid development [5, 6].

Preclinical studies have shown that compared to ADCs with non-cleavable linkers, those with cleavable linkers

likely release free payload prematurely, leading to increased systemic toxicity. In this study, we sought to delineate the potential effect of linker design on systemic toxicity by analyzing the results of clinical trials using ADCs constructed with both types of linkers. We hypothesized that ADCs with cleavable linkers would be associated with greater systemic toxicities than those with non-cleavable linkers. To test this hypothesis, we conducted a systematic review of adverse events (AEs) occurring in cancer patients treated with commercially available ADCs. We then carried out a meta-analysis on all eligible phase II-III clinical trials. We also evaluated the potential effect of drug-to-antibody ratio (DAR) and systemic free payload concentration on toxicity in the context of the cleavability of the linkers used.



**Fig. 1** PRISMA flowchart describing the result of the search and selection process

## 2 Materials and methods

### 2.1 Search methods and study selection

Following the Preferred Reporting Items for Systematic Reviews and Meta-analyses (PRISMA) 2020 updated guidance, two systematic reviews were performed through a PubMed search on July 5, 2022. The first identified studies reporting clinical toxicity rates. Studies eligible for inclusion met the following criteria: (1) clinical trial for cancer therapy in adults ages 18 or older, (2) participants in at least one arm of the study were treated with single-agent ADC, (3) the ADC used was commercially available and FDA-approved for cancer treatment as of July 5, 2022, (4) the study reported treatment-related adverse events, and (5) the study was published in English. Studies where the ADC was administered in combination with other chemotherapy

agents were excluded because the ADC's individual contribution to the overall toxicity profile of the regimen could not be fully delineated. A second systematic review identified studies reporting the DARs and estimated systemic free payload concentrations for all FDA-approved ADCs.

### 2.2 Statistical analysis: meta-analysis of clinical toxicity rates

To compare the incidence of toxicities in patients treated with ADCs constructed with cleavable vs non-cleavable linkers, generalized linear mixed effects logistic models were conducted for each toxicity, including:

1. *By linker type.* Univariable mixed effects logistic regression models were constructed evaluating the association between frequency of each specific toxicity for the binary outcome variables (any grade vs none; grade  $\geq 3$

**Table 1** Included ADCs with linker type, drug-to-antibody ratio, and systemic maximum free payload concentration

| Antibody drug conjugate | Indication(s)                                                                                               | Antigen       | Payload agent | Number of studies | Number of patients | Linker        | Drug-to-antibody ratio (DAR) | Systemic maximum free payload concentration (kg/L) |
|-------------------------|-------------------------------------------------------------------------------------------------------------|---------------|---------------|-------------------|--------------------|---------------|------------------------------|----------------------------------------------------|
| Belantamab mafodotin    | Multiple myeloma                                                                                            | BCMA          | MMAF          | 1                 | 97                 | Non-cleavable | 4                            | 0.0004                                             |
| Brentuximab vedotin     | Hodgkin lymphoma, anaplastic large cell lymphoma, CD30+ peripheral T cell lymphoma, CD30+ mycosis fungoides | CD30          | MMAE          | 11                | 660                | Cleavable     | 4                            | 0.0026                                             |
| Enfortumab vedotin      | Urothelial cancer                                                                                           | Nectin-4      | MMAE          | 3                 | 515                | Cleavable     | 3.8                          | 0.0040                                             |
| Gemtuzumab ozogamicin   | Acute myeloid leukemia                                                                                      | CD33          | Calicheamicin | 2                 | 175                | Cleavable     | 2.5                          | 0.0229                                             |
| Inotuzumab ozogamicin   | Acute lymphoblastic leukemia                                                                                | CD22          | Calicheamicin | 2                 | 205                | Cleavable     | 6                            | 0.0556                                             |
| Loncastuximab tesirine  | Large B-cell lymphoma                                                                                       | CD19          | PBD dimer     | 1                 | 145                | Cleavable     | 2.3                          | 0.0003                                             |
| Moxetumomab pasudotox   | Hairy cell leukemia                                                                                         | CD22          | PE38          | 1                 | 80                 | Cleavable     | 1                            | No systemic accumulation observed                  |
| Sacituzumab govitecan   | Breast cancer<br>Urothelial cancer                                                                          | Trop-2        | SN38          | 3                 | 479                | Cleavable     | 7.6                          | 0.0120                                             |
| Tisotumab vedotin       | Cervical cancer                                                                                             | Tissue factor | MMAE          | 2                 | 156                | Cleavable     | 4                            | 0.0026                                             |
| Trastuzumab deruxtecan  | Breast cancer<br>Gastric cancer                                                                             | HER2          | Deruxtecan    | 3                 | 570                | Cleavable     | 8                            | 0.4347                                             |
| Trastuzumab emtansine   | Breast cancer                                                                                               | HER2          | Emtansine     | 12                | 4,797              | Non-cleavable | 3.5                          | 0.0012                                             |

*BCMA* B-cell maturation antigen, *MMAE* monomethyl auristatin, *MMAF* monomethyl auristatin F, *PBD* pyrrolobenzodiazepine, *PE38* the 38 kDa fragment of *Pseudomonas* exotoxin A, *SN38* active metabolite of irinotecan

**Table 2** Studies included in meta-analysis

| ADC name               | Study first author, year | Cancer type            | Study design/phase/site                        | ADC group sample size | Modified NOS rating |
|------------------------|--------------------------|------------------------|------------------------------------------------|-----------------------|---------------------|
| Belantamab mafadotin   | Lonial, 2021             | Multiple myeloma       | RCT/2/multicenter                              | 97                    | Good                |
| Brentuximab vedotin    | Pro, 2012                | Systemic ALCL          | Single-arm study/2/Multicenter                 | 58                    | Fair                |
|                        | Younes, 2012             | Hodgkin lymphoma       | Single-arm study/2/Multicenter                 | 102                   | Good                |
|                        | Horwitz, 2014            | PTCL                   | Single-arm study/2/Multicenter                 | 35                    | Fair                |
|                        | Monjanel, 2014           | Hodgkin lymphoma, ALCL | Retrospective cohort study/2/Single center     | 45                    | Good                |
|                        | Duvic, 2015              | pcALCL, MF             | Single-arm study/2/Single center               | 54                    | Good                |
|                        | Kim, 2015                | MF, Sezary syndromeS   | Single-arm study/2/Multicenter                 | 32                    | Fair                |
|                        | Prince, 2017             | MF, pcALCL             | RCT/3/Multicenter                              | 64                    | Good                |
|                        | Walewski, 2018           | Hodgkin lymphoma       | Single-arm study/4/Multicenter                 | 60                    | Good                |
|                        | Stefoni, 2020            | Hodgkin lymphoma       | Single-arm study/2/Multicenter                 | 18                    | Good                |
|                        | Kuruvilla, 2021          | Hodgkin lymphoma       | RCT/3/Multicenter                              | 153                   | Good                |
|                        | Song, 2021               | Hodgkin lymphoma, ALCL | Single-arm study/2/Multicenter                 | 39                    | Fair                |
|                        | Rosenberg, 2019          | Urothelial cancer      | Single-arm study/2/Multicenter                 | 125                   | Good                |
|                        | Powles, 2021             | Urothelial cancer      | RCT/3/Multicenter                              | 301                   | Good                |
|                        | Yu, 2021                 | Urothelial cancer      | Single-arm study/2/Multicenter                 | 89                    | Good                |
| Gemtuzumab ozogamicin  | Taksin, 2007             | AML                    | Single-arm study/2/Multicenter                 | 57                    | Fair                |
|                        | Amadori, 2016            | AML                    | RCT/3/Multicenter                              | 118                   | Good                |
| Inotuzumab ozogamicin  | Kantarjian, 2013         | ALL                    | Single-arm study/1.5/Single center             | 41                    | Good                |
|                        | Kantarjian, 2019         | ALL                    | RCT/3/Multicenter                              | 164                   | Good                |
| Loncastuximab tesirine | Caimi, 2021              | DLBCL                  | Single-arm study/2/Multicenter                 | 145                   | Good                |
| Moxetumomab pasudotox  | Kreitman, 2021           | Hairy cell leukemia    | Single-arm study/2/Multicenter                 | 80                    | Good                |
| Sacituzumab govitecan  | Bardia, 2019             | Breast cancer          | Single-arm study/1.5/Multicenter               | 108                   | Good                |
|                        | Bardia, 2021             | Breast cancer          | RCT/3/Multicenter                              | 258                   | Good                |
|                        | Tagawa, 2021             | Urothelial cancer      | Non-randomized multicohort study/2/Multicenter | 113                   | Good                |
| Tisotumab vedotin      | Hong, 2020               | Cervical cancer        | Single-arm study/1.5/Multicenter               | 55                    | Fair                |
|                        | Coleman, 2021            | Cervical cancer        | Single-arm study/2/Multicenter                 | 101                   | Good                |
| Trastuzumab deruxtecan | Modi, 2020               | Breast cancer          | Single-arm study/2/Multicenter                 | 184                   | Good                |
|                        | Shitara, 2020            | Gastric cancer         | RCT/2/Multicenter                              | 125                   | Good                |
|                        | Cortés, 2022             | Breast cancer          | RCT/3/Multicenter                              | 261                   | Good                |
|                        | Burris, 2011             | Breast cancer          | Single-arm study/2/Multicenter                 | 112                   | Good                |
| Trastuzumab emtansine  | Krop, 2012               | Breast cancer          | Single-arm study/2/Multicenter                 | 110                   | Good                |
|                        | Verma, 2012              | Breast cancer          | RCT/3/Multicenter                              | 495                   | Good                |
|                        | Yardley, 2015            | Breast cancer          | Single-arm study/4/Multicenter                 | 215                   | Good                |
|                        | Kashiwaba, 2016          | Breast cancer          | Single-arm study/2/Multicenter                 | 73                    | Good                |
|                        | Krop, 2017               | Breast cancer          | RCT/3/Multicenter                              | 403                   | Good                |
|                        | Watanabe, 2017           | Breast cancer          | Single-arm study/2/Multicenter                 | 232                   | Fair                |
|                        | Montemurro, 2019         | Breast cancer          | Single-arm study/3b/Multicenter                | 2002                  | Good                |
|                        | Von Minckwitz, 2019      | Breast cancer          | RCT/3/Multicenter                              | 743                   | Good                |
|                        | Cortés, 2020             | Breast cancer          | RCT/1.5/Multicenter                            | 80                    | Fair                |
|                        | Emens, 2020              | Breast cancer          | RCT/2/Multicenter                              | 69                    | Good                |
|                        | Cortés, 2022             | Breast cancer          | RCT/3/Multicenter                              | 263                   | Good                |

ALCL anaplastic large cell lymphoma, ALL acute lymphoblastic leukemia, AML acute myeloid leukemia, DLBCL diffuse large B-cell lymphoma, MF mycosis fungoides, NOS Newcastle–Ottawa scale, pcALCL primary cutaneous ALCL, PTCL peripheral T cell lymphoma, RCT randomized controlled trial

versus grade  $\leq 2$ ) and ADC linker type (cleavable vs non-cleavable).

2. *Linker type adjusted for drug-to-antibody ratio.* Multivariable mixed effects logistic regression models were constructed evaluating the association between frequency of each specific toxicity for the binary outcome variables (any grade vs none; grade  $\geq 3$  vs grade  $\leq 2$ ) and ADC linker type (cleavable vs non-cleavable) plus drug-to-antibody ratio (numeric predictor) with the potential interaction between ADC linker type and drug-to-antibody ratio when estimable. The model includes the interaction when estimable since cleavability status and drug-to-antibody ratio are not necessarily independent factors but arise from the design of each medication.

3. *Linker type adjusted for estimated systemic free payload concentration.* Multivariable mixed effects logistic regression models were constructed evaluating the association between frequency of each specific toxicity for the binary outcome variables (any grade vs none; grade  $\geq 3$  vs grade  $\leq 2$ ) and ADC linker type (cleavable vs non-cleavable) plus systemic free payload concentration (numeric predictor) with the potential interaction between ADC linker type and payload systemic free concentration when estimable. Again, the model includes the interaction when

estimable since cleavability status and systemic free payload concentration are not necessarily independent factors but arise from the design of each medication.

Heterogeneity in the estimated probability of each specific toxicity between studies was assessed with the I<sup>2</sup> [2] statistic, describing the percentage of variation in probability of each specific toxicity across the studies arising from differences in the included trials (heterogeneity) rather than sampling error (chance).

### 3 Results

**Study inclusion** A literature search and review of references identified 440 relevant publications after duplicates were removed. After eligibility assessment, a total of 40 clinical trials involving 7,879 patients were used to perform our meta-analysis, as shown in Fig. 1 [7–46]. Eleven (11) commercially-available FDA-approved ADCs were included (Table 1). Nine of these studies reported the results of treatment with ADCs with cleavable linkers ( $N=2,985$ ), whereas two used non-cleavable linkers ( $N=4,894$ ). Table 2 lists the ADC agent, target disease, study design, number of patients

**Table 3** Toxicity  $\geq$  Grade 3 by Cleavability of Linker, Drug-to-Antibody Ratio, and Interaction of Cleavability\*Drug-to-Antibody Ratio

|                                      | # Studies Reporting | # Patients At Risk | # Patients With Toxicity | Cleavability Association (p-value) | Noncleavable = Lower Toxicity | DAR (p-value) | Lower DAR = Less Toxicity | Interaction Cleavability* DAR (p-value) | Interaction Moderating Association | I <sup>2</sup> (%) |
|--------------------------------------|---------------------|--------------------|--------------------------|------------------------------------|-------------------------------|---------------|---------------------------|-----------------------------------------|------------------------------------|--------------------|
| Any Toxicity $\geq$ Grade 3          | 30                  | 7,193              | 3,169                    | 0.002                              | Yes                           | 0.000         | Yes                       | 0.002                                   | Yes                                | 91.895             |
| <b>HEMATOLOGIC</b>                   |                     |                    |                          |                                    |                               |               |                           |                                         |                                    |                    |
| Neutropenia $\geq$ Grade 3           | 30                  | 4,772              | 614                      | 0.021                              | Yes                           | 0.000         | Yes                       | 0.042                                   | Yes                                | 83.002             |
| Leukopenia $\geq$ Grade 3            | 16                  | 2,551              | 173                      | 0.008                              | Yes                           | 0.115         | NS                        | 0.017                                   | Yes                                | 71.442             |
| Anemia $\geq$ Grade 3                | 31                  | 7,185              | 349                      | 0.001                              | Yes                           | 0.027         | Yes                       | 0.001                                   | Yes                                | 77.075             |
| Hematologic $\geq$ Grade 3           | 6                   | 3,975              | 624                      | 0.959                              | NS                            | 0.249         | NS                        | ID                                      | ID                                 | 93.092             |
| Thrombocytopenia $\geq$ Grade 3      | 26                  | 6,522              | 517                      | 0.282                              | NS                            | 0.782         | NS                        | 0.257                                   | NS                                 | 92.705             |
| Lymphopenia $\geq$ Grade 3           | 8                   | 947                | 90                       | 0.612                              | NS                            | 0.274         | NS                        | ID                                      | ID                                 | 44.802             |
| <b>GASTROINTESTINAL</b>              |                     |                    |                          |                                    |                               |               |                           |                                         |                                    |                    |
| Nausea $\geq$ Grade 3                | 32                  | 7,060              | 101                      | 0.557                              | NS                            | 0.000         | Yes                       | 0.556                                   | NS                                 | 1.329              |
| Vomiting $\geq$ Grade 3              | 29                  | 6,653              | 91                       | 0.560                              | NS                            | 0.508         | NS                        | 0.576                                   | NS                                 | 53.511             |
| Diarrhea $\geq$ Grade 3              | 31                  | 6,990              | 120                      | 0.802                              | NS                            | 0.058         | NS                        | 0.874                                   | NS                                 | 62.225             |
| Elevated AST $\geq$ Grade 3          | 25                  | 6,500              | 163                      | 0.525                              | NS                            | 0.239         | NS                        | 0.564                                   | NS                                 | 71.383             |
| Elevated ALT $\geq$ Grade 3          | 21                  | 5,834              | 106                      | 0.911                              | NS                            | 0.266         | NS                        | ID                                      | ID                                 | 57.674             |
| Constipation $\geq$ Grade 3          | 23                  | 5,730              | 20                       | 0.636                              | NS                            | 0.718         | NS                        | 0.637                                   | NS                                 | 0.277              |
| Liver Toxicity $\geq$ Grade 3        | 20                  | 11,508             | 243                      | 0.508                              | NS                            | 0.256         | NS                        | ID                                      | ID                                 | 82.614             |
| <b>CONSTITUTIONAL</b>                |                     |                    |                          |                                    |                               |               |                           |                                         |                                    |                    |
| Pyrexia $\geq$ Grade 3               | 23                  | 5,219              | 37                       | 0.004                              | Yes                           | 0.024         | No                        | 0.006                                   | Yes                                | 46.414             |
| Fatigue $\geq$ Grade 3               | 33                  | 7,352              | 225                      | 0.903                              | NS                            | 0.239         | NS                        | 0.847                                   | NS                                 | 60.924             |
| Headache $\geq$ Grade 3              | 15                  | 4,305              | 30                       | 0.554                              | NS                            | 0.710         | NS                        | 0.552                                   | NS                                 | 35.244             |
| <b>OTHER</b>                         |                     |                    |                          |                                    |                               |               |                           |                                         |                                    |                    |
| Peripheral Neuropathy $\geq$ Grade 3 | 22                  | 2,985              | 107                      | 0.005                              | Yes                           | 0.042         | Yes                       | ID                                      | ID                                 | 84.451             |
| ILD $\geq$ Grade 3                   | 12                  | 3,906              | 21                       | 0.257                              | NS                            | 0.833         | NS                        | ID                                      | ID                                 | 0.000              |
| Ocular Toxicity $\geq$ Grade 3       | 6                   | 807                | 50                       | 0.926                              | NS                            | 0.556         | NS                        | 0.925                                   | NS                                 | 0.000              |
| Hypokalemia $\geq$ Grade 3           | 15                  | 2,593              | 58                       | 0.871                              | NS                            | 0.753         | NS                        | 0.863                                   | NS                                 | 41.073             |

DAR = drug-to-antibody ratio, ID = insufficient data to evaluate interaction, NS = not significant, ILD = interstitial lung disease, ALT = alanine transaminase, AST = aspartate aminotransferase



treated with the ADC, and modified Newcastle–Ottawa scale study quality rating. Table 3 lists the 21 specific toxicities examined. It also indicates the number of included studies reporting the specific toxicity, patients at risk, and the number of patients experiencing toxicities for grade  $\geq 3$  (Table 3) and any grade (Table 4).

As quantified in Tables 3 and 5, at least half the studies reported thrombocytopenia, neutropenia, anemia, increased AST and ALT, nausea, vomiting, diarrhea, and fatigue, as well as any toxicity at any grade and grade  $\geq 3$  AEs. Other toxicities were reported less frequently.

**Systemic toxicity rates with ADCs, stratified by cleavable vs non-cleavable linker** The meta-analytic point estimate of the proportion of patients experiencing each of the 21 toxicities with a 95% confidence interval is displayed in Fig. 2A for toxicities  $\geq$  grade 3, and in Fig. 2B for any grade toxicity, both stratified by linker type. Composite AEs  $\geq$  grade 3 occurred in 43% of patients overall, 47% in the cleavable linker-treated patients and 34% in the non-cleavable-treated patients, and these differences were significant (weighted risk difference  $-12.9\%$ ; 95% CI:  $-17.1\%$  to  $-8.8\%$ ). Specific toxicities  $\geq$  grade 3 with significantly lower proportions favoring non-cleavable linkers were neutropenia ( $-9.1\%$ ; 95% CI  $-12\%$  to  $-6.2\%$ ) and anemia ( $-1.7\%$ ; 95% CI  $-3.3\%$  to  $-0.1\%$ ). There was no significant difference in

rates of grade  $\geq 3$  thrombocytopenia, increased AST/ALT, or fatigue. For all grade toxicities, there were no significant differences in rates of nausea, vomiting, diarrhea, hypokalemia, or headache.

**Linker type and drug-to-antibody ratio** We further examined the potential association between ADC linker type and drug-to-antibody ratios and the estimated probabilities of systemic toxicity. Since the linker type and drug-to-antibody ratio are design features of each ADC and thus are not independent factors, the interaction between linker type and drug-to-antibody ratio was modeled and estimated whenever feasible. A summary of the results for the 21 toxicities are represented as a heatmap (Tables 3 and 4). The p-values are color-coded for level of significance and direction of association.

For grade  $\geq 3$  toxicities (Table 3), non-cleavable linkers remain significantly and independently associated with lower toxicity for any AE ( $p=0.002$ ), neutropenia ( $p=0.021$ ), leukopenia ( $p=0.008$ ), anemia ( $p=0.001$ ), pyrexia ( $p=0.004$ ), and peripheral neuropathy ( $p=0.005$ ) when adjusted for DAR and their interaction where estimable. In addition, higher DAR was significantly and independently associated with higher probability of grade  $\geq 3$  toxicity for any AE, neutropenia, anemia, nausea ( $p<0.001$ ), and peripheral neuropathy. Higher DAR was significantly and independently associated

**Table 4** Toxicity Any Grade by Cleavability of Linker, Drug-to-Antibody Ratio, and Interaction of Cleavability \* Drug-to-Antibody Ratio

|                       | # Studies Reporting | # Patients At Risk | # Patients With Toxicity | Cleavability Association (p-value) | Noncleavable = Lower Toxicity | DAR (p-value) | Lower DAR = Less Toxicity | Interaction Cleavability* DAR (p-value) | Interaction Moderating Association | I <sup>2</sup> (%) |
|-----------------------|---------------------|--------------------|--------------------------|------------------------------------|-------------------------------|---------------|---------------------------|-----------------------------------------|------------------------------------|--------------------|
| Any AE                | 27                  | 6528               | 6173                     | 0.987                              | NS                            | 0.059         | NC                        | 0.992                                   | NS                                 | 91.023             |
| HEMATOLOGIC           |                     |                    |                          |                                    |                               |               |                           |                                         |                                    |                    |
| Neutropenia           | 27                  | 3949               | 916                      | 0.475                              | NS                            | 0.000         | Yes                       | 0.562                                   | NS                                 | 88.8               |
| Leukopenia            | 16                  | 2551               | 375                      | 0.167                              | NS                            | 0.034         | Yes                       | 0.229                                   | NS                                 | 88.888             |
| Anemia                | 28                  | 6812               | 1075                     | 0.111                              | NS                            | 0.002         | Yes                       | 0.133                                   | NS                                 | 89.24              |
| Hematologic           | 6                   | 3975               | 1026                     | 0.691                              | NS                            | 0.008         | Yes                       | ID                                      | ID                                 | 92.09              |
| Thrombocytopenia      | 22                  | 6228               | 1380                     | 0.823                              | NS                            | 0.390         | NS                        | 0.709                                   | NS                                 | 96.593             |
| Lymphopenia           | 7                   | 908                | 125                      | 0.997                              | NS                            | 0.697         | NS                        | ID                                      | ID                                 | 53.794             |
| GASTROINTESTINAL      |                     |                    |                          |                                    |                               |               |                           |                                         |                                    |                    |
| Nausea                | 37                  | 7609               | 2816                     | 0.228                              | NS                            | 0.000         | Yes                       | 0.271                                   | NS                                 | 92.775             |
| Vomiting              | 29                  | 6808               | 1264                     | 0.097                              | NS                            | 0.000         | Yes                       | 0.106                                   | NS                                 | 88.301             |
| Diarrhea              | 34                  | 7301               | 1374                     | 0.811                              | NS                            | 0.013         | Yes                       | 0.766                                   | NS                                 | 92.89              |
| Elevated AST          | 26                  | 6542               | 1172                     | 0.923                              | NS                            | 0.227         | NS                        | 0.935                                   | NS                                 | 94.804             |
| Elevated ALT          | 23                  | 6108               | 846                      | 0.781                              | NS                            | 0.392         | NS                        | ID                                      | ID                                 | 94.345             |
| Constipation          | 24                  | 5945               | 1125                     | 0.366                              | NS                            | 0.182         | NS                        | 0.402                                   | NS                                 | 91.765             |
| Liver (all)           | 26                  | 13084              | 2344                     | 0.925                              | NS                            | 0.240         | NS                        | 0.958                                   | NS                                 | 97.491             |
| CONSTITUTIONAL        |                     |                    |                          |                                    |                               |               |                           |                                         |                                    |                    |
| Pyrexia               | 24                  | 5609               | 989                      | 0.854                              | NS                            | 0.341         | NS                        | 0.808                                   | NS                                 | 82.785             |
| Fatigue               | 34                  | 7584               | 2635                     | 0.091                              | NS                            | 0.385         | NS                        | 0.092                                   | NS                                 | 94.253             |
| Headache              | 15                  | 4546               | 1042                     | 0.016                              | No                            | 0.483         | NS                        | 0.023                                   | Yes                                | 86.633             |
| OTHER                 |                     |                    |                          |                                    |                               |               |                           |                                         |                                    |                    |
| Peripheral neuropathy | 23                  | 3050               | 658                      | 0.054                              | NS                            | 0.358         | NS                        | ID                                      | ID                                 | 93.793             |
| ILD                   | 12                  | 1937               | 78                       | 0.562                              | NS                            | 0.020         | Yes                       | ID                                      | ID                                 | 84.672             |
| Ocular (all)          | 9                   | 1290               | 285                      | 0.008                              | Yes                           | 0.006         | No                        | 0.009                                   | Yes                                | 93.742             |
| Hypokalemia           | 15                  | 2065               | 248                      | 0.280                              | NS                            | 0.553         | NS                        | 0.298                                   | NS                                 | 81.218             |

AE = adverse event, DAR = drug-to-antibody ratio, ID = insufficient data to evaluate interaction, NS = not significant, ILD = interstitial lung disease, ALT = alanine transaminase, AST = aspartate aminotransferase



with lower probability of grade  $\geq 3$  pyrexia. There were significant interaction terms between cleavability status and DAR for any AE ( $p=0.002$ ), neutropenia ( $p=0.042$ ), leukopenia ( $p=0.017$ ), anemia ( $p=0.001$ ), and pyrexia ( $p=0.006$ ). These were all moderating interactions, indicating a lower toxicity then would be predicted from the additive effects of a linear increase in DAR and linker cleavability type.

For any grade toxicity (Table 4), non-cleavable linkers were not significantly and independently associated with toxicity adjusted for DAR and their interaction except lower ocular and higher headache toxicity of any grade. However, higher DARs were significantly and independently associated with a higher probability of any grade toxicity for neutropenia, anemia, leukopenia, all hematological toxicities, nausea, vomiting, diarrhea, and interstitial lung disease. Although the sample size is limited, higher DAR was significantly and independently associated with lower probability of any grade ocular toxicity.

As observed in the heatmap of these data, the direction of the associations when significant were typically in the pre-specified clinically expected direction: higher DAR was associated with higher probability of grade  $\geq 3$  toxicity. As reported in Tables 3 and 4, most of the I [2] statistics

are  $> 50\%$  indicating at least moderate to high levels of heterogeneity in reported probabilities of each specific toxicity between studies.

**Linker type, systemic free payload agent concentration, and toxicity** Next we considered the potential association between ADC linker type and the systemic free payload concentration, including the potential interaction between linker type and systemic free payload concentration when estimable. These results are summarized as a heatmap of the p-values for the significance of the estimated coefficients for each factor in Tables 5 (toxicities  $\geq$  grade 3) and 6 (any grade toxicity). The p-values are color-coded for level of significance and direction of regression coefficient associations between systemic free payload concentrations and linker type. For grade  $\geq 3$  toxicities (Table 5), non-cleavable linkers remain significantly and independently associated with lower toxicity for only peripheral neuropathy ( $p=0.001$ ). Neutropenia, leukopenia, and pyrexia no longer had a significant independent association with linker type after adjustment for the systemic free payload concentration. In fact, the probability of any AE  $\geq$  grade 3 ( $p=0.009$ ) and anemia ( $p=0.011$ ) was higher in patients treated with

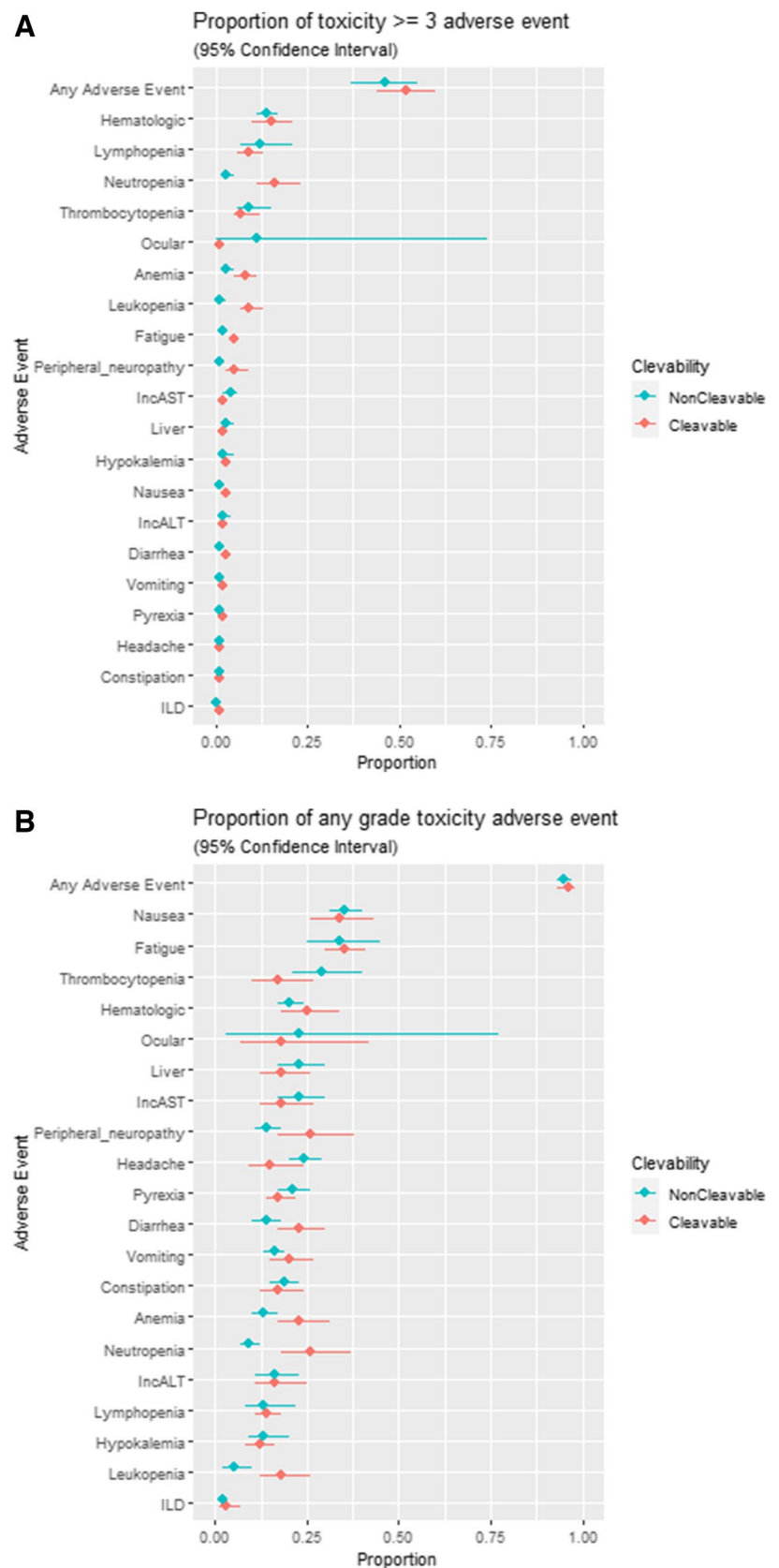
**Table 5** Toxicity  $\geq$  Grade 3 by Cleavability of Linker, Systemic Free Payload Concentration, and Interaction of Cleavability\* Systemic Free Payload Concentration

|                                      | # Studies Reporting | # Patients At Risk | # Patients With Toxicity | Cleavability Association (p-value) | Noncleavable = Lower Toxicity | Free Payload Concentration | Lower Systemic Free Payload Concentration = Less Toxicity | Cleavability* Systemic Free Payload Concentration (p-value) | Interaction Moderating Association | I2 (%) |
|--------------------------------------|---------------------|--------------------|--------------------------|------------------------------------|-------------------------------|----------------------------|-----------------------------------------------------------|-------------------------------------------------------------|------------------------------------|--------|
| Any Toxicity $\geq$ Grade 3          | 29                  | 7113               | 3145                     | 0.009                              | No                            | 0.096                      | NS                                                        | 0.004                                                       | Yes                                | 94.018 |
| <b>HEMATOLOGIC</b>                   |                     |                    |                          |                                    |                               |                            |                                                           |                                                             |                                    |        |
| Neutropenia $\geq$ Grade 3           | 29                  | 4692               | 610                      | 0.792                              | NS                            | 0.08                       | NS                                                        | 0.065                                                       | NS                                 | 89.391 |
| Leukopenia $\geq$ Grade 3            | 15                  | 2471               | 166                      | 0.078                              | NS                            | 0.549                      | NS                                                        | 0.020                                                       | Yes                                | 76.438 |
| Anemia $\geq$ Grade 3                | 30                  | 7105               | 341                      | 0.011                              | No                            | 0.022                      | Yes                                                       | 0.001                                                       | Yes                                | 77.183 |
| Hematologic $\geq$ Grade 3           | 5                   | 3575               | 584                      | 0.688                              | NS                            | 0.986                      | NS                                                        | ID                                                          | ID                                 | 94.512 |
| Thrombocytopenia $\geq$ Grade 3      | 25                  | 6442               | 512                      | 0.167                              | NS                            | 0.72                       | NS                                                        | 0.275                                                       | NS                                 | 93.271 |
| Lymphopenia $\geq$ Grade 3           | 7                   | 867                | 74                       | 0.153                              | NS                            | 0.604                      | NS                                                        | ID                                                          | ID                                 | 10.131 |
| <b>GASTROINTESTINAL</b>              |                     |                    |                          |                                    |                               |                            |                                                           |                                                             |                                    |        |
| Nausea $\geq$ Grade 3                | 31                  | 6980               | 99                       | ID                                 | ID                            | ID                         | ID                                                        | ID                                                          | ID                                 |        |
| Vomiting $\geq$ Grade 3              | 29                  | 6653               | 91                       | 0.738                              | NS                            | 0.897                      | NS                                                        | 0.553                                                       | NS                                 | 53.984 |
| Diarrhea $\geq$ Grade 3              | 30                  | 6910               | 120                      | 0.601                              | NS                            | 0.284                      | NS                                                        | 0.816                                                       | NS                                 | 64.674 |
| Elevated AST $\geq$ Grade 3          | 25                  | 6500               | 163                      | 0.781                              | NS                            | 0.232                      | NS                                                        | 0.513                                                       | NS                                 | 71.765 |
| Elevated ALT $\geq$ Grade 3          | 20                  | 5754               | 105                      | 0.690                              | NS                            | 0.86                       | NS                                                        | ID                                                          | ID                                 | 62.729 |
| Constipation $\geq$ Grade 3          | 22                  | 5650               | 20                       | 0.999                              | NS                            | 0.518                      | NS                                                        | 0.999                                                       | NS                                 | 0.520  |
| Liver Toxicity $\geq$ Grade 3        | 20                  | 11508              | 243                      | 0.276                              | NS                            | 0.582                      | NS                                                        | ID                                                          | ID                                 | 83.777 |
| <b>CONSTITUTIONAL</b>                |                     |                    |                          |                                    |                               |                            |                                                           |                                                             |                                    |        |
| Pyrexia $\geq$ Grade 3               | 22                  | 5139               | 36                       | 0.174                              | NS                            | 0.035                      | Yes                                                       | 0.027                                                       | Yes                                | 60.354 |
| Fatigue $\geq$ Grade 3               | 32                  | 7272               | 225                      | 0.551                              | NS                            | 0.239                      | NS                                                        | 0.883                                                       | NS                                 | 56.331 |
| Headache $\geq$ Grade 3              | 14                  | 4225               | 30                       | 0.908                              | NS                            | 0.954                      | NS                                                        | 0.911                                                       | NS                                 | 34.930 |
| <b>OTHER</b>                         |                     |                    |                          |                                    |                               |                            |                                                           |                                                             |                                    |        |
| Peripheral Neuropathy $\geq$ Grade 3 | 22                  | 2985               | 107                      | 0.000                              | Yes                           | 0.015                      | No                                                        | ID                                                          | ID                                 | 81.545 |
| ILD $\geq$ Grade 3                   | 12                  | 3906               | 21                       | 0.355                              | NS                            | 0.428                      | NS                                                        | ID                                                          | ID                                 | 0.000  |
| Ocular Toxicity $\geq$ Grade 3       | 6                   | 807                | 50                       | 1.000                              | NS                            | 0.519                      | NS                                                        | 1.000                                                       | NS                                 | 0.000  |
| Hypokalemia $\geq$ Grade 3           | 14                  | 2513               | 56                       | 0.903                              | NS                            | 0.624                      | NS                                                        | 0.849                                                       | NS                                 | 43.456 |

ID = insufficient data to evaluate interaction, NS = not significant, ILD = interstitial lung disease, ALT = alanine transaminase, AST = aspartate aminotransferase



**Fig. 2** **A:** Proportion of patients experiencing each of the 21 toxicities  $\geq$  grade 3 with 95% confidence interval. ILD=interstitial lung disease, IncAST=increased aspartate aminotransferase, IncALT=increased alanine aminotransferase. **B:** Proportion of patients experiencing each of the 21 toxicities any grade with 95% confidence interval. ILD=interstitial lung disease, IncAST=increased aspartate aminotransferase, IncALT=increased alanine aminotransferase



non-cleavable linkers when adjusted for systemic free payload concentrations. Similarly, higher systemic free payload concentrations were significantly and independently associated with higher probability of anemia ( $p=0.022$ ) and pyrexia ( $p=0.035$ ). For any grade toxicities (Table 6), non-cleavable linkers remain independently associated with lower peripheral neuropathy ( $p=0.025$ ) but no other toxicity when adjusted for systemic free payload concentration. The main effect of higher systemic free payload concentration was significantly associated with increased probability for any toxicity ( $p=0.043$ ), anemia ( $p=0.028$ ), lymphopenia ( $p=0.005$ ), nausea ( $p<0.001$ ), vomiting ( $p=0.006$ ), constipation ( $p=0.041$ ), and interstitial lung disease ( $p<0.001$ ) after adjustment for linker type.

Again the  $I^2$  statistics tend to be  $>50\%$  indicating at least moderate to high levels of heterogeneity in probabilities of the toxicities between studies.

**ADC linker type, DAR, and free payload concentration** As noted, linker-type, DAR, and systemic free payload concentration are not necessarily independent varying characteristics for each ADC. To further investigate this, we described the relationships between cleavability type,

DAR, and measured systemic free payload concentration for the 11 FDA-approved ADCs under evaluation in this study. These characteristics are summarized in Table 1. Non-cleavable linkers have a numerically lower mean estimated DAR ( $3.75 \pm 0.35$  versus  $4.78 \pm 2.18$ ,  $p=0.544$ ) and lower estimated systemic free payload concentration ( $1.89 \times 10^{-5} \text{ m}^2/\text{L} \pm 1.41 \times 10^{-5} \text{ m}^2/\text{L}$  versus  $1.62 \times 10^{-3} \text{ m}^2/\text{L} \pm 3.64 \times 10^{-3} \text{ m}^2/\text{L}$ ,  $p=0.567$ ). These differences are not statistically different, likely due to the small number of agents being compared (2 non-cleavable and 9-cleavable). However, the higher measured systemic free payload concentrations were significantly associated with higher DARs ( $p=0.043$ ) as depicted in Fig. 3.

## 4 Discussion

In this review and meta-analysis, we sought to delineate how features of ADC design, including linker cleavability, DAR, and systemic free payload concentration, may contribute to their associated systemic toxicities. The results support the hypothesis that ADCs with cleavable linkers are associated with more systemic toxicities than those with non-cleavable

**Table 6** Toxicity Any Grade by Cleavability of Linker, Systemic Free Payload Concentration, and interaction of Cleavability \* Systemic Free Payload Concentration

|                         | # Studies Reporting | # Patients At Risk | # Patients With Toxicity | Cleavability Association (p-value) | Noncleavable = Lower Toxicity | Free payload Agent Concentration | Lower Systemic Free Payload Concentration = Less Toxicity | Cleavability* Systemic Free Payload Concentration (p-value) | Interaction Moderating Association | I <sup>2</sup> (%) |
|-------------------------|---------------------|--------------------|--------------------------|------------------------------------|-------------------------------|----------------------------------|-----------------------------------------------------------|-------------------------------------------------------------|------------------------------------|--------------------|
| Any AE                  | 26                  | 6448               | 6094                     | 0.905                              | NS                            | 0.043                            | Yes                                                       | 0.908                                                       | NS                                 | 91.235             |
| <b>HEMATOLOGIC</b>      |                     |                    |                          |                                    |                               |                                  |                                                           |                                                             |                                    |                    |
| Neutropenia             | 26                  | 3869               | 912                      | 0.762                              | NS                            | 0.051                            | NS                                                        | 0.495                                                       | NS                                 | 92.730             |
| Leukopenia              | 15                  | 2471               | 367                      | 0.993                              | NS                            | 0.078                            | NS                                                        | 0.201                                                       | NS                                 | 89.662             |
| Anemia                  | 27                  | 6732               | 1058                     | 0.385                              | NS                            | 0.028                            | Yes                                                       | 0.129                                                       | NS                                 | 91.408             |
| Hematologic             | 5                   | 3575               | 972                      | 0.559                              | NS                            | 0.127                            | NS                                                        | ID                                                          | ID                                 | 91.962             |
| Thrombocytopenia        | 21                  | 6148               | 1371                     | 0.191                              | NS                            | 0.078                            | NS                                                        | 0.656                                                       | NS                                 | 96.494             |
| Lymphopenia             | 6                   | 828                | 109                      | 0.294                              | NS                            | 0.005                            | Yes                                                       | ID                                                          | ID                                 | 0.000              |
| <b>GASTROINTESTINAL</b> |                     |                    |                          |                                    |                               |                                  |                                                           |                                                             |                                    |                    |
| Nausea                  | 36                  | 7529               | 2788                     | 0.656                              | NS                            | 0.000                            | Yes                                                       | 0.417                                                       | NS                                 | 93.544             |
| Vomiting                | 29                  | 6808               | 1264                     | 0.180                              | NS                            | 0.006                            | Yes                                                       | 0.197                                                       | NS                                 | 90.726             |
| Diarrhea                | 33                  | 7221               | 1357                     | 0.546                              | NS                            | 0.605                            | NS                                                        | 0.877                                                       | NS                                 | 94.425             |
| Elevated AST            | 26                  | 6542               | 1172                     | 0.924                              | NS                            | 0.938                            | NS                                                        | 0.879                                                       | NS                                 | 95.092             |
| Elevated ALT            | 22                  | 6028               | 829                      | 0.880                              | NS                            | 0.757                            | NS                                                        | ID                                                          | ID                                 | 94.772             |
| Constipation            | 23                  | 5865               | 1107                     | 0.755                              | NS                            | 0.041                            | Yes                                                       | 0.435                                                       | NS                                 | 91.433             |
| Liver (all)             | 26                  | 13084              | 2344                     | 0.918                              | NS                            | 0.932                            | NS                                                        | 0.884                                                       | NS                                 | 97.613             |
| <b>CONSTITUTIONAL</b>   |                     |                    |                          |                                    |                               |                                  |                                                           |                                                             |                                    |                    |
| Pyrexia                 | 23                  | 5529               | 964                      | 0.425                              | NS                            | 0.242                            | NS                                                        | 0.842                                                       | NS                                 | 80.858             |
| Fatigue                 | 33                  | 7504               | 2608                     | 0.118                              | NS                            | 0.806                            | NS                                                        | 0.109                                                       | NS                                 | 94.630             |
| Headache                | 14                  | 4466               | 1016                     | 0.244                              | NS                            | 0.099                            | NS                                                        | 0.004                                                       | Yes                                | 72.445             |
| <b>OTHER</b>            |                     |                    |                          |                                    |                               |                                  |                                                           |                                                             |                                    |                    |
| Peripheral neuropathy   | 23                  | 3050               | 658                      | 0.025                              | Yes                           | 0.136                            | NS                                                        | ID                                                          | ID                                 | 93.436             |
| ILD                     | 12                  | 1937               | 78                       | 0.211                              | NS                            | 0.000                            | Yes                                                       | ID                                                          | ID                                 | 0.000              |
| Ocular (all)            | 9                   | 1290               | 285                      | 0.110                              | NS                            | 0.658                            | NS                                                        | 0.066                                                       | NS                                 | 97.013             |
| Hypokalemia             | 14                  | 1985               | 235                      | 0.544                              | NS                            | 0.928                            | NS                                                        | 0.297                                                       | NS                                 | 82.609             |

AE = adverse event, ID = insufficient data to evaluate interaction, NS = not significant, ILD = interstitial lung disease, ALT = alanine transaminase, AST = aspartate aminotransferase

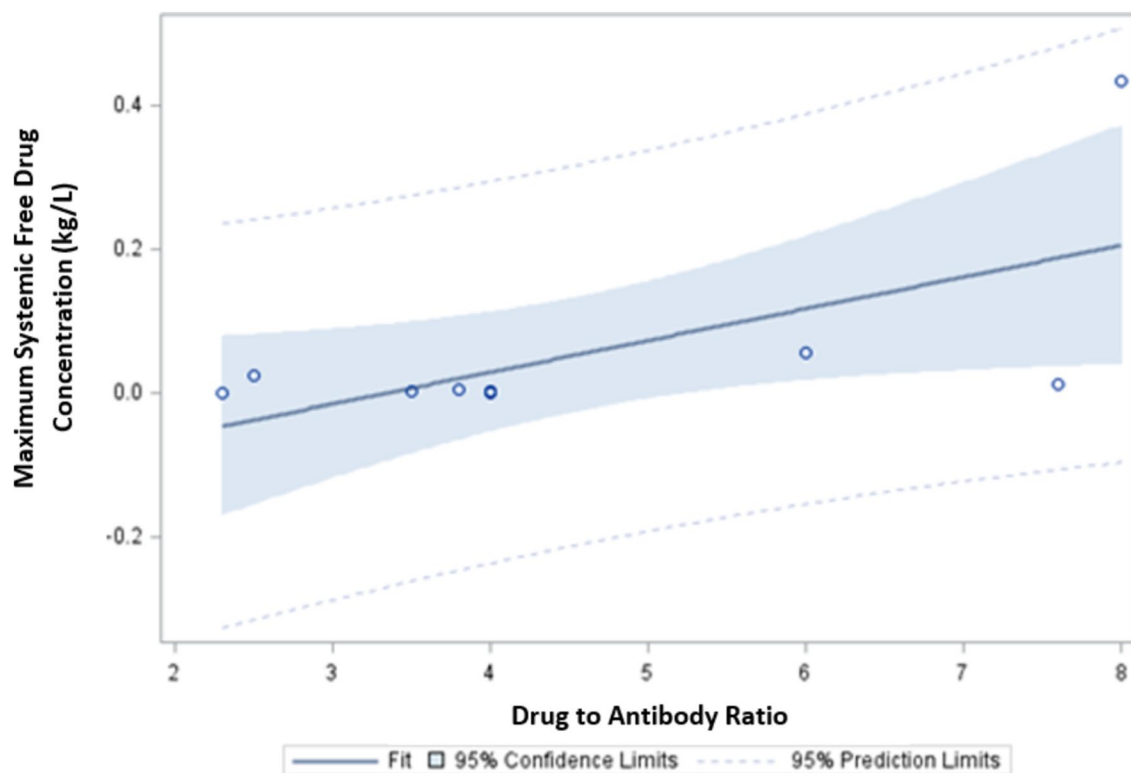


linkers. Interestingly, even though higher DAR was associated with higher grade  $\geq 3$  toxicity, the apparent protective effect of the non-cleavable linker persisted even after adjusting for DAR. However, we found that the association between non-cleavable linkers and lower toxicity was *not* observed after adjusting for systemic free payload concentration. This suggests that systemic free payload concentration is the main factor driving toxicity in ADC-treated patients.

Notably, trastuzumab deruxtecan (T-DXd) has a considerably higher systemic free payload concentration than the other agents studied here. T-DXd has a tetrapeptide cleavable linker that may make it more vulnerable to premature release of its payload. Such a prematurely released payload might explain why T-DXd may be effective in tumor control regardless of HER2 expression. Emerging clinical evidence supports this hypothesis. In a small trial of T-DXd in non-small cell lung cancer patients, the activity of T-DXd was shown to be independent of HER2 over-expression in HER3+, 2+, or 1+ tumors [47]. More recently, clinical trial data presented during the 2021 San Antonio Breast Cancer Symposium showed that T-DXd was active in breast cancer patients regardless of HER2 expression, including HER2 0 tumors [48].

By contrast, trastuzumab emtansine (T-DM1), another anti-HER2 ADC with a non-cleavable linker, has not been shown to have activity in patients with low HER2 tumor expression. This may partly explain why in the DESTINY-03 trial [35], T-DM1 (HER2 dependent) was shown to have lower efficacy compared to T-DXd (HER2 dependent and independent). Consonant with these observations, T-DM1 was also shown to have much lower systemic toxicity compared to T-DXd, likely because of the relative stability of the T-DM1 linker. We've previously proposed that the off-target effects observed with T-DM1 may be the result of payload released from lysed tumor cells [2]. It is tempting to speculate that for these reasons, ADCs with cleavable linkers may in general have higher anti-tumor efficacy albeit higher systemic toxicities.

Our results arise from a meta-analysis using multiple agents used to treat highly disparate patient populations with different malignancies. We suspect this explains most of the heterogeneity observed in probabilities of specific toxicities between studies reflected in the  $I^2$  [2] statistics. Nonetheless, differences in ADC chemical design are shown to be significantly associated with specific clinical toxicities despite this tremendous heterogeneity.



**Fig. 3** Association between DAR and maximum systemic free payload concentration

## 5 Conclusion

In summary, ADCs are rapidly becoming the standard of care for patients across disease sites. The results here show that linker choice and the potential for premature payload release among ADCs can affect their systemic toxicity and efficacy. It will, therefore, be critical in the design of future ADCs to find the appropriate balance between the highest potential efficacy and associated systemic toxicities [49]. In this regard, contemporary studies are focused on the development of novel ADC linkers that can be released conditionally within the tumor microenvironment to increase both the specificity of drug delivery and anti-tumor efficacy. The ideal ADC design should aim for high therapeutic index that balances off-target toxicities, taking into consideration factors such as linker cleavability, DAR, and payload membrane permeability [50]. The results presented here suggest one critical consideration in achieving this balance during future ADC development for cancer patients will be linker design and the potential for premature payload release.

**Author Contribution** S.C.T. conceived of the idea and supervised the study design and manuscript writing. C.W., M.M., and R.P. assisted with data collection and manuscript writing. T.L., Y.Z., and W.H. designed the computational framework and analyzed the data. W.H. also assisted with manuscript writing. N.M. helped supervise the manuscript.

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## Declarations

**Conflict of Interest** The authors declare no competing interests.

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